

ZARA Sales Statistical Analysis and Linear Regression Revenue Model

Gabriela Nirmal

Khoury College of Computer Sciences, Northeastern University - Boston

Introduction

This project presents a statistical analysis of product-level revenue data from ZARA, a leading global fashion brand. Utilizing a dataset from a web scraping snapshot in February 2024, the goal is to identify key product features that influence revenue and to train a predictive model based on those features. The statistical analysis explores the effects of product type, section (men's or women's), promotional status, seasonality, price, and store placement on total revenue. A linear regression model is trained and evaluated to predict total revenue, offering insights into what drives higher sales and how this information can be used to better understand trends in product performance.

Methods

Approach

This project consists of two main parts: a statistical analysis of categorical and numerical features within the dataset, and a regression modeling task to predict total revenue. The analysis includes exploratory data visualizations and summary statistics to understand revenue trends across categories. Feature engineering was also applied by calculating revenue as the product of

price and sales volume, followed by preprocessing techniques such as MinMax scaling for numerical features and one-hot encoding for categorical features.

Data

The data used in this analysis was obtained from "ZARA Sales" on Kaggle and contains information on 252 ZARA products, scraped on February 2, 2024. Although the size is relatively small and not representative of ZARA's complete product line, it provides a useful subset for exploratory modeling. Key numerical features include price and sales volume, from which total revenue was calculated. Important categorical variables include:

- **Position:** store placement (e.g., front of store, end cap, aisle)
- **Promotion:** whether the product is on promotion
- **Seasonal:** whether the product is part of a seasonal collection
- **Section:** men's or women's department
- **Terms:** product type (e.g., jacket, shoes, sweater)

To prepare the data for regression, MinMax scaling was applied to price, and categorical variables were transformed with one-hot encoding.

Pre-Training Statistical Analysis

Before modeling, I conducted a statistical analysis to explore trends and distributions in the dataset. I began by visualizing the distribution of numerical features, specifically price and sales volume. The price distribution resembled a Gaussian distribution, with most items falling below \$100. Sales volume, on the other hand, was uniformly distributed with no particular concentration in any range.

I also analyzed categorical data by graphing the number of items in each category, such as the count of items by section (men's or women's), promotion status, seasonal vs. non-seasonal, store position, and product type (terms). This gave a better idea of how the product types and marketing strategies were distributed across the dataset.

Next, I compared average values, particularly total revenue, within each category. For example, I looked at the average revenue of promoted vs. non-promoted items, or jackets vs. t-shirts. This revealed some early insights into which product types and marketing strategies would likely be more successful. Notably, men's clothing showed a slightly higher average revenue overall, but further modeling revealed that women's clothing had stronger effects once other factors were taken into consideration.

Training

Two linear models were trained and compared: a ridge regression model and a closed-form linear regression model. Both models were trained on the same scaled and encoded dataset, with total revenue as the target variable.

- **Ridge Regression MSE: 0.0124, R^2 Score: 0.5457**
- **Closed-Form Linear Regression MSE: 0.0116, R^2 Score: 0.5726**

While the difference in mean squared error appears small, in a scaled context, this difference can reflect more meaningful performance changes. The closed-form model also had a noticeably higher R^2 score, suggesting it captured more variance in the data. Therefore, the closed-form model was chosen for the final interpretation of the data.

Evaluation

The model was evaluated using mean squared error (MSE), R^2 score, and comparison between predicted and actual revenue values from 10 randomly selected products. To interpret feature importance, weights from the linear model were examined.

Key results:

- **Price** had a weight of **0.9968**, making it the most influential feature by far. This makes sense since revenue is calculated as price multiplied by sales volume. Price is one of the two fundamental components of revenue.
- **Promotion (Yes)** had a positive weight of **0.0063**, indicating that promotional products tended to generate slightly higher revenue.
- **Seasonal (Yes)** had a positive weight of **0.0025**.
- **Women's Section** had a notable positive impact (**0.0248**).
- **Jackets** had a more positive impact on revenue than other categories like shoes, jeans, sweaters, or t-shirts.
- **Aisle placement** performed better than end cap or front-of-store positions.
- Product types like **t-shirts** and **Shoes** had **negative weights**, indicating lower revenue contributions relative to other categories like jackets.

Interestingly, the average revenue by category showed that men's clothing initially appeared to generate more revenue than women's. However, after training the model and accounting for other factors (like product type, seasonality, and promotion), the weights indicated that women's clothing actually contributes more to revenue overall. This shows the importance of considering multiple variables and their interactions rather than relying on raw averages alone.

Results

The model's high weight on price confirms that pricing is the dominant driver of revenue in this dataset. Other features like promotion status, seasonality, section, and product type also had a measurable influence. Jackets stood out as having more revenue-driving potential than items like shoes, jeans, sweaters, or t-shirts. Placement in the aisle section of the store outperformed front-of-store or end cap positioning. Women's clothing also emerged as a stronger revenue contributor once other product features were considered.

Conclusions

While the linear model provided useful insights, there are limitations due to the dataset's small size and single-day snapshot nature. A broader dataset covering more features like time, product variety, and store locations would improve generalizability.

From this analysis, we can conclude:

- **Price** overwhelmingly drives revenue, which makes sense since it is one of the two components that make up total revenue.
- **Promotion, seasonality, store placement, and product category** all play meaningful roles in shaping revenue.
- **Women's products** tend to generate higher revenue once all factors are considered.
- **Jackets** outperformed categories like shoes and t-shirts, and **aisle** placement tended to do better than front-of-store or end cap.

These results help highlight key patterns in product performance and how different attributes influence revenue. Future work could involve more complex models like decision trees or polynomial regression, or experimenting with interaction terms like price and promotion together to explore more nuanced effects.